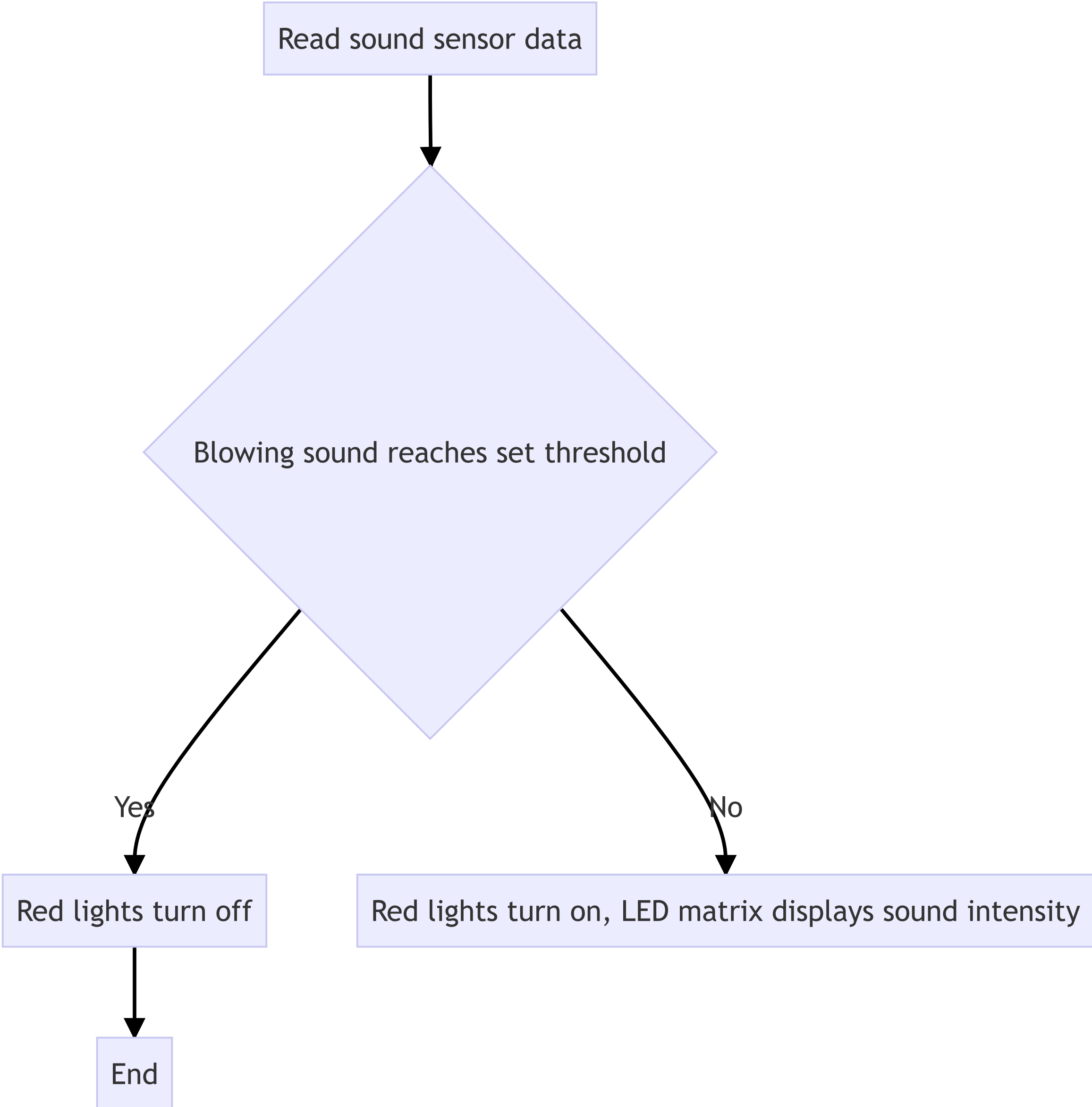
4. Blowing Candles

Function Description

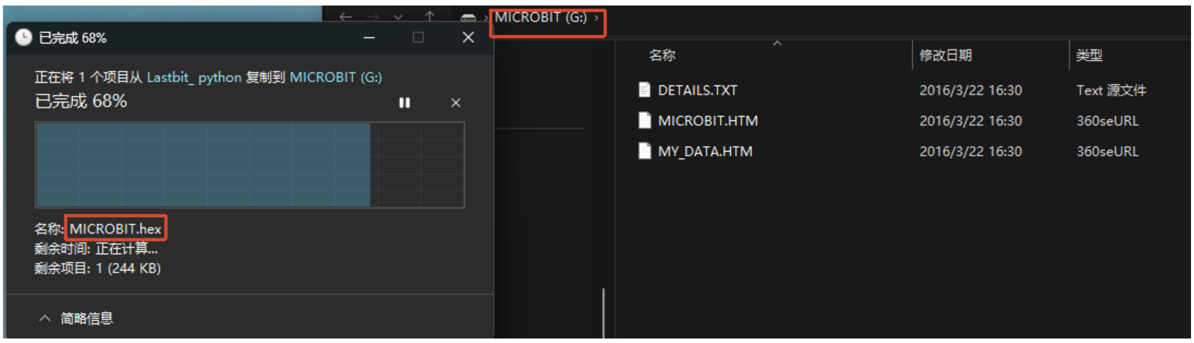
In this section, we will learn to use the microphone sensor to detect sound intensity and achieve the "Blowing Candles" effect.

Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



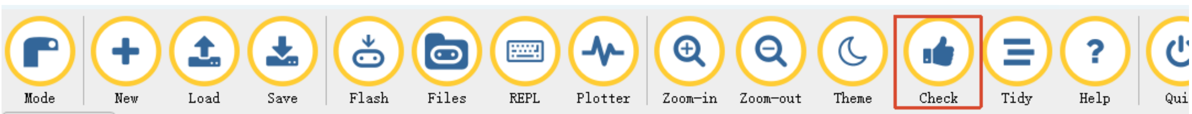
1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.

   Click the `New` button in the toolbar at the top left, and a file titled "untitled" will appear.

3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

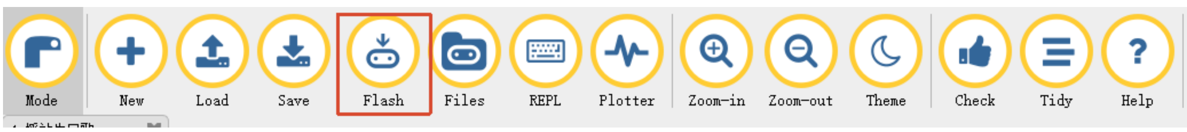
4. Now you can click `Check` to verify whether there are any problems with the code format. This case involves multi-line display patterns that may trigger warnings, but this does not affect flashing or running!



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows `Copied code onto micro:bit.` at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.



6. Unplug the microUSB data cable and switch the switch to the ON position.

Code Implementation

Method 1: Use the built-in microphone of the micro:bit

```
# 4. Blowing Candles
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

# Sound level
item = 0
# First run flag
frist_run = 0

# Initialize: Show candle pattern, set red lights
Lastbit.all_lights_set_color(255, 0, 0)
display.show(Image("00900:"
                  "00900:"
                  "00900:"
                  "00000:"
                  "00900"))
sleep(2000)
display.clear()
while True:
    if frist_run == 0:
        # Read sound level
        item = microphone.sound_level()

        # Show different flame sizes based on sound level
        if item > 40 and item < 70:
            display.show(Image("00000:"
                              "00000:"
                              "00000:"
                              "00900:"
                              "99999"))

        elif item >= 70 and item < 110:
            display.show(Image("00000:"
                              "00000:"
                              "00000:"
                              "09990:"
                              "99999"))

        elif item >= 110 and item < 160:
            display.show(Image("00000:"
                              "00000:"
                              "00900:"
                              "99999:"
                              "99999"))

        elif item >= 160 and item < 170:
            display.show(Image("00000:"
                              "00000:"
                              "99999:"
                              "99999:"
                              "99999"))

        elif item >= 170 and item < 200:
```

```
            display.show(Image("00000:"
                               "09990:"
                               "99999:"
                               "99999:"
                               "99999"))

        elif item >= 200 and item < 240:
            display.show(Image("00900:"
                               "99999:"
                               "99999:"
                               "99999:"
                               "99999"))

        elif item >= 240:
            # Max volume reached, turn off lights and show smiley
            display.show(Image("99999:"
                               "99999:"
                               "99999:"
                               "99999:"
                               "99999"))
            Lastbit.all_lights_off()
            display.clear()
            frist_run = 1
            display.show(Image("09990:"
                               "90009:"
                               "00000:"
                               "09090:"
                               "00000"))

    sleep(100)
```

Method 2: Use the MIC on the expansion board

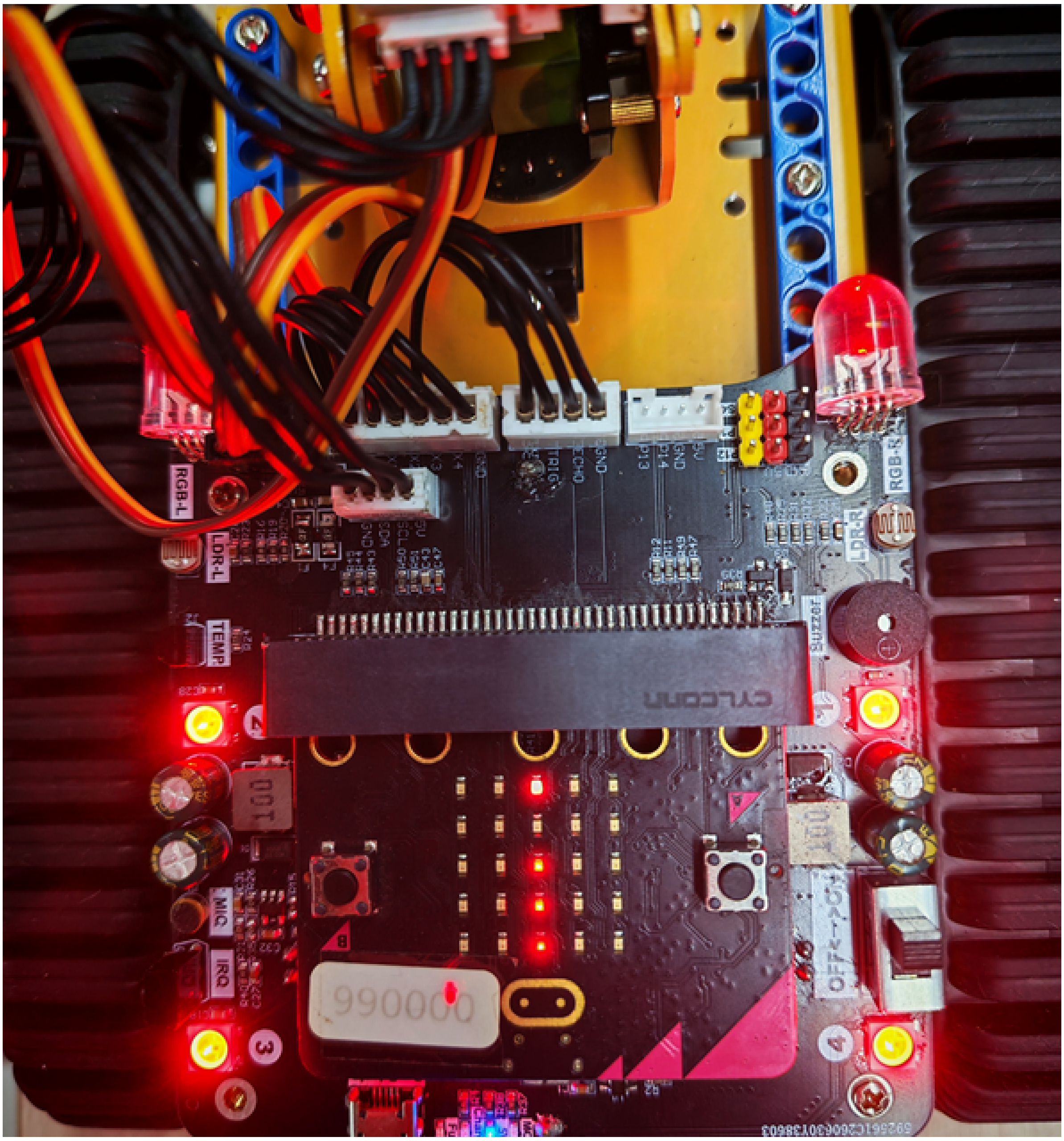
```
# 4. Blowing Candles (Expansion board MIC)
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

# Sound level
item = 0
# First run flag
frist_run = 0

# Initialize: Show candle pattern, set red lights
Lastbit.all_lights_set_color(255, 0, 0)
display.show(Image("00900:"
                   "00900:"
                   "00900:"
                   "00000:"
                   "00900"))
sleep(2000)
display.clear()
while True:
    if frist_run == 0:
        # Read sound level
        item = Lastbit.sound_detect()[2]
```

```python
        # Show different flame sizes based on sound level
        if item > 40 and item < 70:
            display.show(Image("00000:"
                               "00000:"
                               "00000:"
                               "00900:"
                               "99999"))

        elif item >= 70 and item < 110:
            display.show(Image("00000:"
                               "00000:"
                               "00000:"
                               "09990:"
                               "99999"))

        elif item >= 110 and item < 160:
            display.show(Image("00000:"
                               "00000:"
                               "00900:"
                               "99999:"
                               "99999"))

        elif item >= 160 and item < 170:
            display.show(Image("00000:"
                               "00000:"
                               "99999:"
                               "99999:"
                               "99999"))

        elif item >= 170 and item < 200:
            display.show(Image("00000:"
                               "09990:"
                               "99999:"
                               "99999:"
                               "99999"))

        elif item >= 200 and item < 240:
            display.show(Image("00900:"
                               "99999:"
                               "99999:"
                               "99999:"
                               "99999"))

        elif item >= 240:
            # Max volume reached, turn off lights and show smiley
            display.show(Image("99999:"
                               "99999:"
                               "99999:"
                               "99999:"
                               "99999"))
            Lastbit.all_lights_off()
            display.clear()
            frist_run = 1
            display.show(Image("09990:"
                               "90009:"
                               "00000:"
                               "09090:"
```

```
                                        "00000"))
    sleep(100)
```

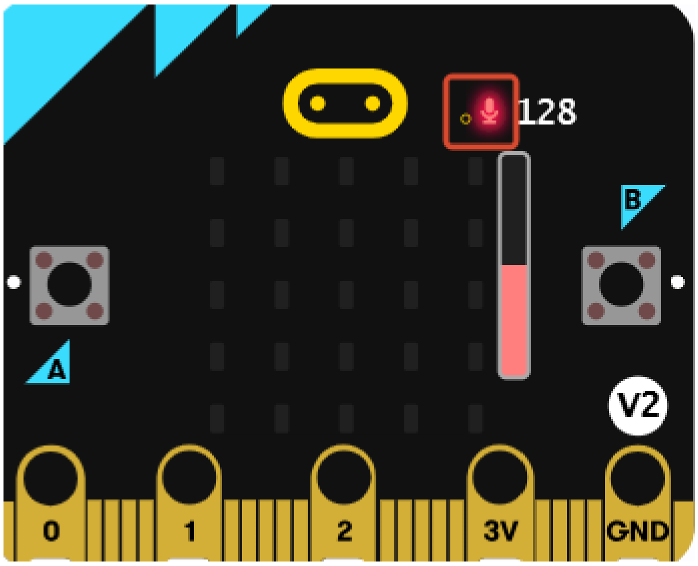
Experimental Phenomenon

After flashing is complete and the power is turned on, the car will first light up red lights, and a red candle pattern will be displayed on the LED matrix screen.

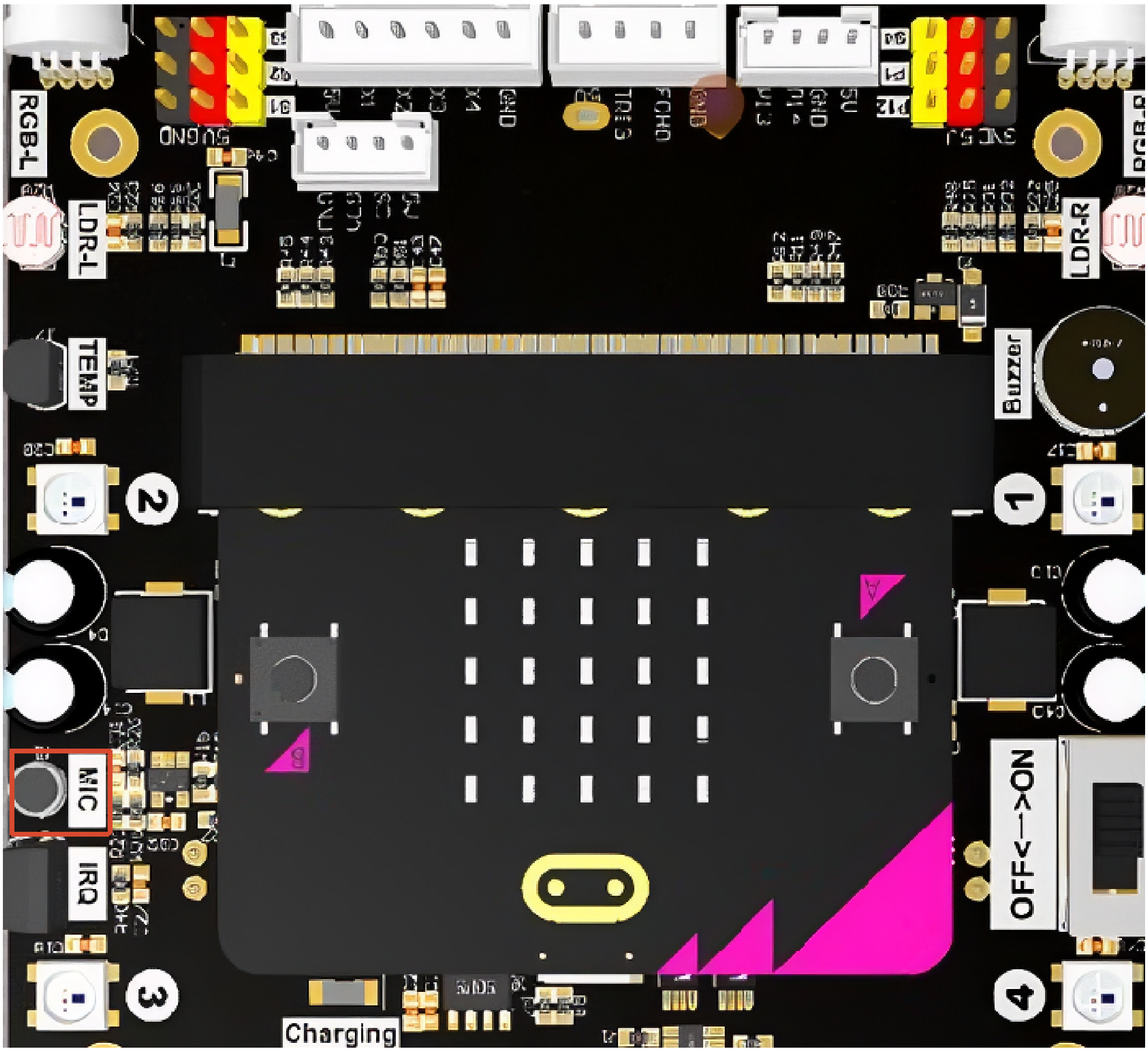


After the red candle pattern disappears, blow air toward the microphone. When all LEDs on the matrix screen light up, the car lights turn off and a smiley face is displayed, simulating the candle being blown out.

The recording position of the built-in microphone of the micro:bit is as follows:



The recording position of the MIC on the expansion board is as follows:



Notes

- When using the built-in microphone of the micro:bit, use `microphone.sound_level()` to read the sound
- When using the MIC on the expansion board, use `Lastbit.sound_detect()[2]` to read the sound
- The sound detection threshold can be adjusted according to the actual environment